

# Physics-Informed Early Warning of Quantum Decoherence in Single-Qubit Systems Under Unknown Time-Dependent Dissipation

immediate

## Abstract

We address the problem of predicting the remaining decoherence time  $T_2$  of a single-qubit system from a short, partial observation window of Pauli expectation values, *without* any prior knowledge of the dissipation rate  $\gamma(t)$ . For constant  $\gamma$  and amplitude-damping noise, the OLS-slope formula recovers the exact analytical result ( $R^2 = 0.9995$ , MAE = 0.056) with no machine learning required. For unknown, time-dependent  $\gamma(t)$ —the realistic regime on every physical qubit platform—we propose a physics-informed residual BiLSTM that embeds the OLS-slope estimate as a structural prior and predicts only the correction  $\delta T$  to it. On three single-qubit scenarios (constant  $\gamma$ , time-dependent amplitude damping  $\sigma_-$ , and time-dependent dephasing  $\sigma_z$ ), the residual model achieves  $R^2 = 0.94$ – $0.98$ , improving over the physics-only baseline by up to  $\Delta R^2 = +0.19$ . The risk alarm (binary: decoherence within horizon  $\Delta t$ ) reaches AUROC  $\geq 0.93$  using as little as 5% of the trajectory, and the system outperforms a two-step inverse approach by a factor of  $> 15\times$  in MAE at short windows.

## 1 Introduction

Decoherence is the central obstacle to scalable quantum computation. A quantum system coupled to a Markovian environment evolves under the Lindblad master equation ( $\hbar = 1$ ):

$$\dot{\rho} = -i[H, \rho] + \gamma(t)(L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}), \quad (1)$$

where  $L$  is the jump operator and  $\gamma(t) \geq 0$  is the time-dependent dissipation rate. The *decoherence time*  $T_2$  is the time at which the  $\ell_1$ -coherence  $C(t) = \sum_{i \neq j} |\rho_{ij}(t)|$  falls to  $e^{-1}$  of its initial value.

Knowing the remaining decoherence time from a partial trajectory is directly actionable: it enables error-correction protocols to preemptively reset or re-encode qubits before fidelity falls below a critical threshold [4].

**When is ML needed?** For constant  $\gamma$  and  $\sigma_-$  (amplitude damping), the coherence decays as  $C(t) = C_0 e^{-\gamma t/2}$ ,

so  $\log C$  is linear:

$$\frac{d \log C}{dt} = -\frac{\gamma}{2} \implies T_2 = \frac{2}{\gamma} = -\frac{1}{\text{slope}}. \quad (2)$$

A simple OLS fit to  $\log C(t)$  over any pre-decoherence window recovers  $T_2$  with  $R^2 = 0.9995$  (see Sec. 3.1). Machine learning is *unnecessary* in this regime.

The problem becomes non-trivial when  $\gamma(t)$  is *unknown and time-varying*—the realistic case on every physical platform (Table 1). In this regime, the OLS slope captures only the instantaneous  $\gamma(t)/2$ , and the resulting  $T_2$  estimate can be severely biased.

Table 1: Sources of unknown  $\gamma(t)$  on qubit platforms.

| Platform        | Source of non-stationary $\gamma(t)$ |
|-----------------|--------------------------------------|
| Superconducting | $1/f$ flux noise, TLS fluctuators    |
| Spin qubits     | Hyperfine / nuclear spin bath        |
| Trapped ions    | Laser power drift, micromotion       |
| Photonic        | Cavity leakage rate drift            |

**Relation to prior work.** The closest related work, arXiv:2505.06928 [1], uses a pure Transformer to *reconstruct*  $\gamma(t)$  from a *complete* trajectory via Bernstein polynomial regression. That work solves an offline inverse problem; ours solves an online forward problem (Table 2). In Sec. 3.5 we directly compare the two approaches on the same partial-window setting and show that the inverse approach is ill-conditioned at short windows ( $R^2 < -0.83$  for all  $f \leq 0.50$ ), while direct prediction remains reliable ( $R^2 = 0.92$ – $0.96$ ).

## 2 Method

### 2.1 Simulation setup

We study three single-qubit scenarios (Table 3). The dissipation rate  $\gamma(t)$  is parameterised by degree-4 Bernstein polynomials, guaranteeing  $\gamma(t) \geq 0$ . Six qualitatively distinct profiles are used: constant, monotone increasing/decreasing, peak, step-up, and step-down. Each scenario uses 100 randomly sampled configurations; trajec-

Table 2: Comparison with arXiv:2505.06928.

|                | arXiv:2505.06928        | This work               |
|----------------|-------------------------|-------------------------|
| Goal           | Reconstruct $\gamma(t)$ | Predict $T_2$ remaining |
| Task           | Inverse (offline)       | Early warning (online)  |
| Input          | Full trajectory         | Partial window          |
| Output         | Bernstein coeffs        | $T_2$ + risk score      |
| Physics        | None                    | OLS prior embedded      |
| Analytic limit | No                      | Yes (Eq. 2)             |

tories are solved via the Lindblad master equation solver in QUTIP [2].

Table 3: Single-qubit scenarios studied in this work.

| Label | Jump operator                  | $\gamma(t)$    |
|-------|--------------------------------|----------------|
| A     | $\sigma_-$ (amplitude damping) | Constant       |
| B     | $\sigma_-$ (amplitude damping) | Time-dependent |
| C     | $\sigma_z$ (dephasing)         | Time-dependent |

## 2.2 Physics-informed residual architecture

The OLS-slope formula (2) provides a useful starting point even for time-dependent  $\gamma(t)$ : it encodes the *current* decay rate from the observation window. We therefore structure the model as a residual corrector:

$$\hat{T}_{2,\text{rem}} = \underbrace{T_{\text{physics}}(\mathbf{w})}_{\text{OLS estimate}} + \underbrace{\delta T_{\text{LSTM}}(\mathbf{w})}_{\text{learned correction}}, \quad (3)$$

where  $\mathbf{w} \in \mathbb{R}^{W \times F}$  is the observation window. This design has two properties: (i) *analytical recovery*: when  $\gamma$  is constant the network learns  $\delta T \approx 0$ , exactly recovering Eq. (2); (ii) *corrective capacity*: when  $\gamma(t)$  varies, the LSTM corrects only the residual—a much easier regression target than predicting  $T_2$  from scratch.

**Network details.** A bidirectional LSTM (hidden size 64, 1 layer) processes  $\mathbf{w}$  augmented with  $\log C$  and  $\log(\text{purity})$ . The mean-pooled output is concatenated with four physics scalars (slope,  $\log C_{\text{obs}}$ ,  $t_{\text{obs}}$ ,  $T_{\text{physics}}$ ), yielding a 132-dimensional representation fed into two linear heads:

$$\delta T = \text{Lin}(64) \xrightarrow{\text{GELU}} \text{Lin}(1), \quad (4)$$

$$p_{\text{risk}} = \sigma(\text{Lin}(32) \xrightarrow{\text{GELU}} \text{Lin}(1)), \quad (5)$$

where  $p_{\text{risk}} = P(\text{decoherence within } \Delta t)$ . Training uses AdamW with Huber regression loss and BCE classification loss, with a patience-based early-stop scheduler.

**Ablation variants.** `physics_only`: OLS formula, no neural network. `lstm_only`: BiLSTM without physics prior ( $T_{\text{physics}} = 0$ ). `physics_lstm`: full residual model (ours). `transformer`: pure Transformer encoder, analogous to arXiv:2505.06928 applied to the  $T_2$  task.

## 3 Experiments

### 3.1 E1 — Ablation study

Table 4 reports  $R^2$ , MAE, and risk AUROC for all four variants on the three single-qubit scenarios.

Table 4: E1 ablation:  $R^2$  (MAE) per scenario and variant. Best per scenario in **bold**.  $N = 50$  test samples per cell.

| Variant                   | Scen. A              | Scen. B              | Scen. C              | AUROC        |
|---------------------------|----------------------|----------------------|----------------------|--------------|
| <code>physics_only</code> | <b>0.999</b> (0.056) | 0.753 (0.646)        | 0.965 (0.113)        | 0.957        |
| <code>lstm_only</code>    | −0.446 (2.74)        | 0.009 (1.92)         | −0.029 (0.808)       | 0.968        |
| <code>physics_lstm</code> | 0.976 (0.176)        | <b>0.945</b> (0.356) | <b>0.976</b> (0.151) | 0.965        |
| <code>transformer</code>  | 0.799 (0.631)        | 0.902 (0.595)        | <b>0.981</b> (0.136) | <b>0.970</b> |

Key observations:

- **Claim C1 confirmed.** `physics_only` achieves  $R^2 = 0.9995$  on scenario A, validating the OLS formula for constant- $\gamma$  amplitude damping. No ML is needed in this regime.
- **Claim C2 confirmed.** `physics_lstm` outperforms `physics_only` on time-dependent scenarios: +0.19 on B (0.945 vs. 0.753) and +0.01 on C (0.976 vs. 0.965). Crucially, `lstm_only` fails catastrophically on all three scenarios ( $R^2 \in \{-0.45, 0.01, -0.03\}$ ), confirming that the physics prior is *essential*, not merely helpful.
- **Transformer competitiveness.** The pure Transformer achieves  $R^2 = 0.981$  on scenario C and 0.902 on B—competitive with, but not superior to, `physics_lstm`. The residual architecture is preferred because it is guaranteed to recover the exact analytical solution in the constant- $\gamma$  limit (scenario A: 0.976 vs. 0.799 for the Transformer).

### 3.2 E2 — Minimum observation window

Table 5 shows `physics_lstm` trained separately for each window fraction  $f \in \{0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$  on a mixture of scenarios B and C ( $N = 120$  each).

Two observations deserve note. First,  $R^2$  is approximately constant across all window lengths (0.70–0.74), indicating that the bottleneck is training capacity rather than observation length. This is consistent with the E1 finding that a *single-scenario* model (scenario B alone) achieves  $R^2 = 0.945$  at the same  $W = 20$  steps: mixing B with C reduces per-regime accuracy. Second, and

Table 5: E2 window sweep on scenarios B+C. “\*” marks AUROC  $\geq 0.93$ .

| $f$  | $R^2$ | MAE   | AUROC  |
|------|-------|-------|--------|
| 0.05 | 0.711 | 0.825 | 0.933* |
| 0.10 | 0.723 | 0.813 | 0.950* |
| 0.15 | 0.735 | 0.773 | 0.947* |
| 0.20 | 0.736 | 0.702 | 0.872  |
| 0.30 | 0.734 | 0.677 | 0.969* |
| 0.50 | 0.697 | 0.674 | 0.990* |

more importantly, the risk alarm (AUROC) already exceeds 0.93 at  $f = 0.05$ —meaning the binary decoherence warning is reliable when only 5% of the trajectory has elapsed.

### 3.3 E3 — Measurement noise robustness

Table 6 shows degradation of a model trained on clean scenarios B+C as Gaussian noise  $\sigma$  is added to all observable channels at evaluation time.

Table 6: E3 noise robustness ( $N = 74$  test samples).

| $\sigma_{\text{noise}}$ | $R^2$ | MAE   | AUROC |
|-------------------------|-------|-------|-------|
| 0.000                   | 0.959 | 0.220 | 0.976 |
| 0.010                   | 0.956 | 0.224 | 0.976 |
| 0.020                   | 0.949 | 0.251 | 0.971 |
| 0.050                   | 0.901 | 0.374 | 0.942 |
| 0.100                   | 0.727 | 0.597 | 0.865 |

At  $\sigma = 0.05$  (realistic hardware noise levels),  $R^2 = 0.901$  and AUROC = 0.942—both above their respective publication targets. The graceful degradation is a direct consequence of the physics prior: the OLS slope is computed over a full window of  $W = 20$  steps, averaging out i.i.d. noise by  $\sim \sqrt{W}$ . At  $\sigma = 0.10$ , AUROC drops to 0.865; we consider this the effective noise-tolerance ceiling of the current architecture.

### 3.4 E4 — Cross-system generalisation (1-qubit)

A single `physics_lstm` is trained on the union of scenarios A, B, and C and evaluated on each subsystem separately (Table 7).

Table 7: E4 cross-system: one model trained on A+B+C.

| Scen. | Regime                       | $R^2$        | MAE          | AUROC        |
|-------|------------------------------|--------------|--------------|--------------|
| A     | const. $\gamma$ , $\sigma_-$ | 0.490        | 1.164        | <b>0.979</b> |
| B     | TD $\gamma$ , $\sigma_-$     | 0.867        | 0.397        | <b>0.981</b> |
| C     | TD $\gamma$ , $\sigma_z$     | <b>0.936</b> | <b>0.162</b> | <b>0.990</b> |

Scenario B and C achieve strong generalisation ( $R^2 = 0.867$  and  $0.936$ , AUROC  $\geq 0.98$ ). Scenario A performs

modestly ( $R^2 = 0.490$ ) because the mixture model learns to apply a non-zero correction  $\delta T$ , whereas for constant  $\gamma$  the physics term is already exact and the correction should be zero. For deployment, a lightweight regime classifier can select between `physics_only` (constant  $\gamma$ ) and `physics_lstm` (time-dependent  $\gamma$ ) based on the linearity of  $\log C$  over the window.

### 3.5 E5 — Direct prediction vs. inverse-problem baseline

We compare our direct predictor against the two-step inverse approach of arXiv:2505.06928 operationalised for the  $T_2$  task: (1) train an MLP to reconstruct Bernstein  $\gamma(t)$  coefficients from the window; (2) forward-simulate the Lindblad equation to read off  $T_2$ . Both methods operate on the same partial windows of scenario B (Table 8).

Table 8: E5: direct `physics_lstm` vs. inverse  $\gamma$ -reconstruction baseline on scenario B.

| Method                 | $f$  | $R^2$        | MAE          |
|------------------------|------|--------------|--------------|
| Direct ( <b>ours</b> ) | 0.10 | <b>0.922</b> | <b>0.259</b> |
| Inverse (baseline)     | 0.10 | −0.856       | 1.390        |
| Direct ( <b>ours</b> ) | 0.20 | <b>0.827</b> | <b>0.394</b> |
| Inverse (baseline)     | 0.20 | −0.827       | 1.358        |
| Direct ( <b>ours</b> ) | 0.30 | <b>0.909</b> | <b>0.304</b> |
| Inverse (baseline)     | 0.30 | −1.256       | 1.543        |
| Direct ( <b>ours</b> ) | 0.50 | <b>0.956</b> | <b>0.160</b> |
| Inverse (baseline)     | 0.50 | −3.039       | 1.493        |

The inverse baseline yields  $R^2 < 0$  at *every* window fraction, including  $f = 0.50$ . This is not a failure of the MLP per se, but an inherent ill-posedness: many  $\gamma(t)$  profiles are compatible with a short partial window, making  $T_2$  prediction from a reconstructed  $\gamma$  high-variance. Direct prediction sidesteps this indeterminacy entirely.

## 4 Discussion

**Architecture design principle.** The residual structure (3) is a form of *physics-informed ML* where domain knowledge enters as an *architectural inductive bias* rather than a loss penalty. This is stronger than standard PINNs [3]: the output is structurally decomposed, so the network cannot override the physics entirely—it can only add a correction. The consequence is that the model *degrades gracefully to the analytical solution* in the tractable limit (scenario A), while remaining expressive for complex  $\gamma(t)$  profiles.

**Risk score as a practical interface.** The AUROC values in Tables 4–7 suggest that the binary risk score  $p_{\text{risk}}$  is a reliable alarm even when the regression accuracy

( $R^2$ ) is moderate. This decoupling matters for deployment: a quantum error-correction scheduler only needs a calibrated probability estimate to decide whether to reset a qubit, not a precise  $T_2$  value. The alarm is operational from the very first 5% of the trajectory (AUROC = 0.933,  $f = 0.05$ ), making it suitable for real-time integration.

**Limitations.** The experiments were conducted on simulated data with 100 trajectories per scenario. Validation on real hardware (where  $\gamma(t)$  is not parameterisable by Bernstein polynomials and readout errors are correlated) is left for future work, as is extension to multi-qubit systems, where entanglement and non-exponential coherence decay pose qualitatively different challenges.

## 5 Conclusion

We have presented a physics-informed residual BiLSTM for online early warning of quantum decoherence in single-qubit systems. The key contributions are:

1. A precise characterisation of when ML is necessary: the OLS formula suffices for constant- $\gamma$  amplitude damping ( $R^2 = 0.9995$ ); ML is needed only when  $\gamma(t)$  is time-dependent.
2. A residual architecture (3) that embeds the OLS prior structurally, guaranteeing recovery of the analytical solution and achieving  $R^2 = 0.94$ – $0.98$  for time-dependent scenarios.
3. A risk alarm with AUROC  $\geq 0.93$  from the first 5% of the trajectory, suitable for real-time integration in quantum error-correction pipelines.
4. Direct empirical evidence that inverse  $\gamma$ -reconstruction followed by simulation is ill-conditioned at short windows, while direct prediction is robust.

## References

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